Data Visualization & Analytics Lab

Experiment No. : 06

Implementation of multi linear regression using Python

Experiment No.06

1. **Aim:** Implement multiple/multivariate linear regression using Python.

2. **Objectives:**
   - Build a model that predicts **multiple dependent variables**.
   - Visualize and analyze relationships in data.

3. **Course Outcomes**: Students will be able to apply and analyse the descriptive data analytics technique **for prediction**

4. **Hardware / Software Required:** Python

5. **Theory:**
   Regression is a task of predicting continuous quantity. Linear regression is a statistical approach for modeling relationship between a dependent variable with a given set of independent variables.
   Multiple Linear Regression is a statistical technique used to model the relationship between one dependent variable and two or more independent variables.

   General form:

$$Y = b_0 + b_1X_1 + b_2X_2 + ... + b_nX_n$$

   Multivariate Regression is a statistical technique used to predict more than one dependent variable using one or more independent variables.
   General form:

$$Y_1, Y_2 = b_0 + b_1X_1 + b_2X_2 + ... + b_nX_n$$

Where:

- **Y**= Dependent variable
- $\mathbf{X_1, X_2 \ldots X_n}$ = Independent variables
- $\mathbf{b_0}$ = Intercept
- $\mathbf{b_1, b_2 \ldots b_n}$ = Regression coefficients

Steps:

1. Import required libraries (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn).
2. Load the dataset.
3. Define independent and dependent variables.
4. Split the dataset into training and testing data.
5. Train the regression model.
6. Predict output variables.
7. Visualize relationships between variables using plots.
8. Evaluate performance using metrics like MAE,RMSE, R-squared Score (R^2)..

Sample Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Load dataset
data = pd.read_csv("data.csv")

# Independent variables
X = data[['X1','X2']]

# Dependent variables
Y = data[['Y1','Y2']]

# Split dataset
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2)
# Create model
```

```
model = LinearRegression()

# Train model
model.fit(X_train, Y_train)

# Predictions
predictions = model.predict(X_test)

print(predictions)

# Visualization: pairplot
sns.pairplot(data)
plt.show()

# Visualization: correlation heatmap
sns.heatmap(data.corr(), annot=True, cmap="coolwarm")
plt.show()
```

6. Conclusion:

We have learnt the use of various libraries of python to implement multi linear regression model. The visualization helped in understanding the correlation between multiple independent and dependent variables in an effective way.

7. Viva Questions:

- What is multiple/multivariate regression?
- Which python libraries are used to build and visualize a regression model?

References:

1. Visualization with Matplotlib: https://www.oreilly.com/library/view/python-data-science/9781491912126/ch04.html
2. https://www.kaggle.com/code/emineyetm/multiple-linear-regression-in-python